Swiss Glaciers: Visualizing Retreat Across Space, Time, and Measurement Type

COM-480 Data Visualization — Milestone 3 Process Book
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Project: Interactive visualization of Swiss glacier retreat using GLAMOS glacier inventories, length-change measurements, and flow-velocity data.

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1. Motivation and Final Goal

Swiss glaciers are one of the clearest visible indicators of long-term environmental change. Their retreat is not only a scientific fact but also a spatial and visual phenomenon: glacier fronts move, outlines shrink, flow patterns change, and different regions evolve at different speeds. However, most available glacier data is presented through expert-oriented tools or scientific tables. These sources are accurate and powerful, but they are not designed for a user who wants to quickly understand the overall story.

Our project aims to make Swiss glacier retreat readable at a glance, while still allowing users to inspect detailed data when they want to go deeper. Instead of presenting every available measurement at once, we designed an interactive website that begins with a clean national map and gradually reveals more information through interaction.

The final visualization is built around three complementary questions:

1. **Where is glacier retreat happening?**
   A national map shows historical glacier outlines in geographic context.
2. **How has each glacier changed over time?**
   A timeline slider and glacier-specific detail pages show area and length-change evolution.
3. **How do glaciers move where velocity data is available?**
   Flow-velocity vectors add a dynamic layer, showing glaciers not only as shrinking shapes but as moving bodies of ice.

The goal is not to replace scientific glacier monitoring platforms. Those tools remain essential for researchers and domain experts. Our goal is different: to create a public-facing explanatory visualization where the main pattern is understandable before the user has to read technical labels or interpret dense overlays.

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2. From Initial Idea to Final Scope

Our initial idea was to build an interactive data story about Swiss glacier retreat. From the beginning, we wanted the project to be map-centered because glacier retreat is inherently spatial. A chart alone can show that a glacier is shrinking, but a map shows where this is happening, how glaciers are distributed across Switzerland, and how retreat differs between regions.

The original concept had three main components:

- a national overview of Swiss glacier outlines,
- a temporal control to compare different historical states,
- selected glacier pages for more detailed exploration.

During the first stages of the project, we also considered several optional extensions. These included rankings of the most affected glaciers, external contextual overlays such as tourism or light pollution, and a more advanced integration of flow-velocity data. As the project evolved, we decided to prioritize the core glacier data instead of adding weakly connected external correlations. This was an important design decision: adding more datasets would have made the project broader, but not necessarily clearer or more trustworthy.

A major difference between our process and a traditional design workflow is that we did not rely heavily on static wireframes. Because the project is interaction-heavy, we treated runnable prototypes as our design sketches. Early versions of the map, the glacier detail pages, and the velocity view helped us test what types of interaction were actually useful. This gave us a more realistic understanding of the design than paper sketches would have, because many of the important questions were interactive: how the map should update, how much information a popup should contain, and whether additional layers would clarify or clutter the view.

The final scope became more focused:

- a national glacier-outline map using historical inventory years,
- a time slider for temporal comparison,
- clickable glaciers with local summaries,
- detail pages with D3-based length-change charts,
- an optional velocity layer where data is available,
- a clean public-facing narrative rather than a dense expert dashboard.

This final direction keeps the original ambition of showing glacier retreat across Switzerland, but avoids pretending that every possible glacier metric can be shown equally well everywhere.

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3. Data Sources and Preprocessing

The project uses several glacier-related datasets, each describing a different aspect of glacier change.

The first dataset is the Swiss Glacier Inventory data from GLAMOS. These historical inventories provide glacier outlines for selected benchmark years: **1850, 1931, 1973, 2010, and 2016**. These outlines are the basis of the national map. They allow users to compare the spatial footprint of glaciers over a long historical period.

The second dataset is glacier length-change data. While glacier outlines describe area and spatial footprint, length-change measurements describe how the glacier front advances or retreats over time. We use this data in the glacier-specific detail pages, where D3 charts show cumulative and periodic length change for selected glaciers such as Aletsch, Gorner, and Rhone.

The third dataset is flow-velocity data. This data describes how fast and in which direction ice moves at measured points. It is not available for every glacier or every year, so we use it as an optional layer rather than as the main visualization. Where available, velocity vectors help users understand glaciers as dynamic systems instead of static polygons.

The basemap provides geographic context. It helps users understand where glaciers are located, how they relate to valleys and mountain regions, and why a spatial interface is useful for this topic.

Preparing the data was one of the most difficult parts of the project. The raw glacier inventory data is not immediately usable in a browser visualization. The shapefiles had to be converted into GeoJSON, simplified into a format that could be loaded by Leaflet, and enriched with properties that the frontend could use for interaction. The preprocessing script groups glacier polygons by glacier identifier, computes or preserves area information, joins against a glacier list, and exports one GeoJSON file per inventory year.

An important limitation is that the current map does not show every raw glacier polygon in the original inventories. It uses a matched subset of named glaciers that can be compared more consistently across the selected years. This decision improves interpretability, but it also means that the map should not be described as a complete inventory of every Swiss glacier polygon. Glacier matching across historical inventories is difficult because glaciers can split, merge, disappear, change identifiers, or lack consistent naming. For a public-facing interactive visualization, a stable matched subset is more understandable than a larger but less consistent raw dataset.

This data limitation shaped the final design. Instead of claiming complete coverage, we emphasize that the visualization combines several trustworthy views of glacier retreat: historical outlines for selected inventory years, detailed length-change series for selected glaciers, and velocity measurements where available.

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4. Main Challenges and Design Tradeoffs

The largest challenge was not simply displaying the data. It was deciding how to display it without misleading the user.

4.1 Matching glaciers over time

A naive version of the project would treat each glacier as a stable object that exists unchanged across all years. In practice, this is not true. Glaciers can split into smaller glaciers, merge into larger systems, disappear, or be represented differently across inventories. This makes long-term comparison difficult. If we simply plotted everything, the map would show more data, but the user might incorrectly interpret unmatched or structurally different polygons as clean temporal changes.

To address this, the final map uses a more controlled matched subset. This is a compromise: it reduces total coverage, but it makes the temporal comparison more coherent.

4.2 Combining metrics with different meanings

Area, length change, mass balance, and velocity are related, but they are not the same measurement.

- **Area** shows the spatial footprint of a glacier.
- **Length change** shows advance or retreat of the glacier front.
- **Velocity** shows how ice moves across the glacier surface.
- **Mass balance** describes annual gain or loss in water equivalent where available.

A key design challenge was to avoid mixing these metrics as if they were interchangeable. The final project therefore separates them visually: area and outlines dominate the map, length change appears in detail charts, and velocity appears as an optional overlay.

4.3 Avoiding visual overload

The official GLAMOS viewer is scientifically rich and useful for expert users, but it also illustrates the design problem we wanted to solve. A first-time user is confronted with many colored outlines, technical measurements, interface controls, and dense visual layers. This is powerful if the user already knows what they are looking for, but it creates a high cognitive load for public communication.

Our visualization takes the opposite approach. The first view is intentionally simple. Users start with a clean map and a clear temporal interaction. More information appears only when they ask for it by clicking a glacier, opening a detail page, or enabling an additional layer.

This principle of progressive disclosure became one of the central design decisions of the project. We wanted the visualization to be simple at first glance, but not shallow.

4.4 Dropping external correlation layers

During planning, we considered adding external overlays such as tourism, infrastructure, or light pollution. We eventually dropped this idea. These layers might have made the project look richer, but they would have introduced a risk of weak or misleading correlations. Since the main challenge was already to make glacier retreat clear and trustworthy, we decided that adding external factors would distract from the core story.

This was a useful constraint. It forced us to improve the main glacier visualization instead of expanding sideways into less reliable claims.

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5. Final Visualization and User Journey

The final website is structured as a layered exploration. The user can begin with a broad national view and then move toward more detailed evidence.

5.1 Landing page

The landing page introduces the project and frames the main question: how Swiss glaciers have changed over time. It acts as the entry point to the map and to selected glacier stories. The goal is to avoid overwhelming the user immediately with raw controls or technical terminology.

5.2 National glacier map

The national map is the core of the visualization. It displays glacier outlines on a geographic basemap and allows users to move through the historical inventory years. The year slider is intentionally simple: rather than pretending that we have continuous area data for every year, it reflects the discrete inventory years available in the dataset.

When the user changes the year, the map updates to show the corresponding glacier outlines. This makes long-term retreat visible as a spatial phenomenon. Instead of first reading a number, users can see the shape and extent of glaciers change directly.

5.3 Glacier selection and popup summaries

Clicking a glacier reveals local information. The popup or side panel gives the glacier name, area information for the selected inventory year, and a compact view of its historical evolution. This interaction turns the map from a passive background into an exploratory tool.

The popup is deliberately limited. It provides enough information for quick inspection, but it does not try to contain every chart and every metric. More detailed information is reserved for the glacier-specific pages.

5.4 Glacier-specific detail pages

The detail pages provide a deeper temporal view for selected glaciers. These pages use D3.js charts to show cumulative and periodic length change. This is where users can inspect a glacier more precisely than on the national map.

The separation between map and detail page is important. The map is best for spatial comparison; the detail page is better for time-series reading. Keeping these tasks separate makes the interface clearer.

5.5 Velocity layer

The velocity layer adds a third perspective. While area outlines show how glacier footprints changed, and length charts show retreat over time, velocity vectors show motion. Where velocity measurements are available, arrows represent the direction and magnitude of ice movement.

This layer is optional because velocity data is not available uniformly across all glaciers. Making it optional prevents the map from becoming cluttered and avoids implying complete coverage. It also gives the user a natural progression: first understand where glaciers are and how they changed, then explore how they move.

Together, these components form a coherent user journey: overview first, details second, advanced layers last.

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6. Visual Encoding and Interaction Design

The visual design follows a simple rule: make the first interpretation easy, then allow deeper inspection.

6.1 Map-centered layout

We chose a map-centered layout because glacier retreat is spatial. The position, shape, and clustering of glaciers matter. A national chart could summarize area loss, but it would hide where retreat happens. The map lets users connect data to geography immediately.

6.2 Temporal slider

The timeline slider is one of the main interactions. It maps directly to the available inventory years and makes temporal comparison intuitive. We chose not to interpolate between inventory years because that could suggest a level of temporal precision the data does not provide.

6.3 Color and hierarchy

The glacier palette uses cold tones associated with ice and snow. More alarming colors, such as red, are reserved for severe change or highlighted states. This creates a visual hierarchy: most of the interface remains calm and readable, while important warnings can still stand out.

This differs from dense expert viewers, where many colors may be meaningful but hard to parse for a general audience. In our design, colors are used sparingly so that the user can understand the map before reading a legend in detail.

6.4 Progressive disclosure

The interface uses progressive disclosure across several levels:

1. The map shows the broad spatial pattern.
2. The slider reveals historical change.
3. Clicking a glacier reveals local metrics.
4. Detail pages show longer time-series charts.
5. The velocity layer adds motion where data is available.

This design avoids the common problem of showing too many variables at once. The user is not forced to understand area, length, velocity, and mass balance simultaneously. Instead, each interaction adds one layer of complexity.

6.5 D3 charts for detailed temporal reading

The glacier detail pages use D3.js because time-series charts require more precise interaction than the map. D3 allows zooming, panning, and custom chart behavior. These charts complement the map: the map provides spatial intuition, while the charts provide temporal detail.

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7. Technical Implementation

The project is implemented as a static web application. This architecture was chosen because it is simple to deploy, reproducible, and sufficient for the interaction patterns we needed. The final website can be hosted directly from the repository, with data stored as static GeoJSON and CSV files.

The main technologies are:

- **HTML/CSS/JavaScript** for the website structure and interaction logic,
- **Leaflet** for the interactive map,
- **D3.js** for glacier-specific charts,
- **GeoJSON** for glacier outlines and spatial features,
- **CSV** for length-change time series,
- **Python** preprocessing scripts for converting raw glacier data into browser-ready files.

The repository is organized around the final website. The `docs/` directory contains the deployed site. The national map is implemented separately from the glacier detail pages, and the data files are stored in a dedicated data directory. This makes the project easy to run locally and compatible with GitHub Pages-style deployment.

The preprocessing step converts raw glacier inventory shapefiles into GeoJSON files for the selected inventory years. Each file can then be loaded dynamically by the frontend. When the user moves the year slider, the map updates the displayed glacier layer. Click events attach glacier metadata to interactive popups, allowing the user to inspect local information.

The glacier-specific pages load length-change CSVs and render them as D3 charts. These charts provide a more precise temporal view than the map can offer. They are especially useful because glacier retreat is not always smooth: a long-term downward trend can include periods of slower retreat or temporary advance.

The velocity data is rendered as an additional map layer. Each measurement point contains direction and magnitude information, which can be represented visually using arrows. This adds a dynamic component to the otherwise outline-based map.

The main technical tradeoff is performance versus completeness. Full glacier inventories can be large and complex, especially if all polygons and all years are included at high resolution. For this milestone, we chose a lightweight static architecture and a controlled dataset that supports smooth interaction in the browser. Future versions could optimize the spatial data further or support more complete raw inventories with additional preprocessing.

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8. Contributions, Limitations, and Future Work

8.1 Team contributions

The project involved both technical implementation and communication work. Since the final milestone is evaluated not only on code but also on the process book, screencast, visual design, and clarity of explanation, the work was divided across implementation, data preparation, prototyping, and narrative development.

Team member	Main contributions
Stefan	Project direction, milestone/report writing, narrative framing, design scope, coordination, presentation structure, validation of the final story and process-book.
Timaël	Early prototypes, feature exploration, interaction ideas, experimental components, and map/velocity feature concepts.
Corentin	Data wrangling, GLAMOS preprocessing, conversion and cleaning support, and preparation of usable datasets for map/chart components.
Lilian	Final implementation, website integration and frontend polishing.

8.2 Limitations

The current project has several limitations.

First, the national map uses a matched subset of named glaciers rather than every raw glacier polygon from the original inventories. This improves interpretability but limits completeness. A fully comprehensive version would need more careful handling of glacier splits, merges, disappearances, and changing identifiers.

Second, area data is available only for selected inventory years. The timeline slider therefore represents discrete historical snapshots, not a continuous annual record.

Third, velocity data is sparse and only available for selected glaciers and years. It is useful as an additional layer, but it should not be interpreted as a complete national velocity dataset.

Fourth, different glacier metrics describe different phenomena. Area, length change, velocity, and mass balance are related but not interchangeable. The visualization tries to keep these distinctions clear, but users may still need guidance when comparing metrics.

Finally, some originally planned contextual overlays were dropped. This made the project narrower, but also more trustworthy. We preferred a clean glacier-focused visualization over a broader project with weaker or less validated correlations.

8.3 Future work

Future work could extend the project in several directions.

A first improvement would be to include more of the raw inventory data while explicitly handling glacier splits and merges. This would allow a more complete national view without sacrificing interpretability.

A second improvement would be to add data availability indicators. For example, the interface could show where area data, length-change data, velocity data, or mass-balance data are available, instead of letting users assume that all metrics exist everywhere.

A third improvement would be a guided story mode. The current website supports exploration, but a future version could guide users through a fixed narrative: national retreat, regional differences, selected glacier case studies, and velocity patterns.

A fourth improvement would be comparison and ranking views. Users could explore which glaciers lost the most area in absolute terms, which lost the largest relative share, or which had the strongest measured retreat over a selected period.

Finally, contextual overlays such as tourism or infrastructure could be reconsidered, but only if they are introduced carefully and with strong explanation. The main lesson from this milestone is that more data is not automatically better. For this topic, clarity and trustworthiness matter more than visual density.

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9. Conclusion

This project evolved from a broad idea — visualizing Swiss glacier retreat — into a layered interactive website combining spatial, temporal, and dynamic glacier data. The final visualization uses a national map for first-glance understanding, selected glacier pages for detailed time-series analysis, and velocity vectors for an additional view of glacier motion.

The main design challenge was balancing richness with readability. Glacier datasets are complex, and expert tools often expose that complexity directly. Our approach was to structure the information progressively: start with the spatial pattern, reveal local details through interaction, and add advanced metrics only when they help the user understand more.

The final result is not a complete scientific monitoring platform. It is a public-facing explanatory visualization that turns glacier retreat into a visual story: simple enough to understand quickly, but grounded enough to support deeper exploration.